Lab Sheet 04

Title: Update, Delete Commands in MongoDB.

Objective:

- Understand about Update operations.
- Understand delete commands.

Dataset: Student Records

StudentID	Name	Age	Grade	Section	Marks (%)	EnrollmentDate	Status
101	Emma Johnson	14	9	A	85	2021-06-10	Active
102	Liam Smith	15	10	B	90	2020-09-12	Active
103	Olivia Brown	13	8	A	88	2022-02-15	Active
104	Noah Wilson	16	11	C	75	2019-07-20	Graduated
105	Ava Miller	14	9	B	92	2021-01-18	Active
106	James White	17	12	C	80	2018-05-25	Graduated
107	Charlotte Lee	15	10	A	87	2020-08-30	Active
108	Ethan Clark	16	11	B	78	2019-09-10	Active
109	Mia Walker	13	8	C	89	2022-03-11	Active
110	Henry Adams	17	12	A	82	2018-04-05	Graduated

- Creates database **SchoolDB** • Creates the collection as **"Students**"
- Insert all data.

```
use SchoolDB

db.createCollection("Students")

db.Students.insertMany([
{StudentID:101, Name:"Emma Johnson", Age:14, Grade:9, Section:"A", Marks:85,
EnrollmentDate:new Date("2021-06-10"), Status:"Active"},
{StudentID:102, Name:"Liam Smith", Age:15, Grade:10, Section:"B", Marks:90,
EnrollmentDate:new Date("2020-09-12"), Status:"Active"},
```

```
{StudentID:103, Name:"Olivia Brown", Age:13, Grade:8, Section:"A", Marks:88,
EnrollmentDate:new Date("2022-02-15"), Status:"Active"},
{StudentID:104, Name:"Noah Wilson", Age:16, Grade:11, Section:"C", Marks:75,
EnrollmentDate:new Date("2019-07-20"), Status:"Graduated"},
{StudentID:105, Name:"Ava Miller", Age:14, Grade:9, Section:"B", Marks:92,
EnrollmentDate:new Date("2021-01-18"), Status:"Active"},
{StudentID:106, Name:"James White", Age:17, Grade:12, Section:"C", Marks:80,
EnrollmentDate:new Date("2018-05-25"), Status:"Graduated"},
{StudentID:107, Name:"Charlotte Lee", Age:15, Grade:10, Section:"A", Marks:87,
EnrollmentDate:new Date("2020-08-30"), Status:"Active"},
{StudentID:108, Name:"Ethan Clark", Age:16, Grade:11, Section:"B", Marks:78,
EnrollmentDate:new Date("2019-09-10"), Status:"Active"},
{StudentID:109, Name:"Mia Walker", Age:13, Grade:8, Section:"C", Marks:89,
EnrollmentDate:new Date("2022-03-11"), Status:"Active"},
{StudentID:110, Name:"Henry Adams", Age:17, Grade:12, Section:"A", Marks:82,
EnrollmentDate:new Date("2018-04-05"), Status:"Graduated"}
])

db.Students.find()
```

Update Operations

1. Increase the marks of all students in **Grade 10** by 5%.

```
db.Students.updateMany(
    {Grade:10},
    {$mul:{Markd:1.05}}
)
```

2. Change the **status** of students in **Grade 12** to "**Alumni**".

```
db.Students.updateMany(
    {Grade:12},
    {$set:{Status:"Alumni"}}
)
```

3. Rename the "**Marks (%)**" field to "**Score**" in all records.

```
db.Students.updateMany(
    {},
    {$rename:{"Marks":"Score"}}
)
```

4. Update the **section** of "**Emma Johnson**" from "**A**" to "**B**".

```
db.Students.updateOne(
    {Name:"Emma Johnson"},
    {$set:{Section:"B"}}
)
```

5. Add a new field "**Scholarship**" for students who scored **above 90%**, with a value of "**Yes**".

```
db.Students.updateMany(

    {Score:{$gt:90}},
    {$set:{Scholarship:"Yes"}}
)
```

Delete Operations

1. Delete students who have the status

```
db.Students.deleteMany(
    {Status:"Graduated"}
)
```

2. Remove students from **Grade 11** who enrolled **before 2020**.

```
db.Students.deleteMany(
  {
    Grade: 11,
    EnrollmentDate: { $lt: new Date("2020-01-01") }
  }
)
```

3. Delete a specific student record for "**James White**".

```
db.Students.deleteMany(
  {Name:"James White"}
)
```

4. Remove the "**Scholarship**" field from all students.

```
db.Students.deleteMany(
  {},
  {$unset:{Scholarship:""}}
)
```

Exercise 01

Dataset: Books Collection

BookID	Title	Author	Genre	Published Year	Copies Available	Total Copies	Status
301	The Great Gatsby	F. Scott Fitzgerald	Fiction	1925	3	5	Available
302	To Kill a Mockingbird	Harper Lee	Fiction	1960	1	4	Available
303	1984	George Orwell	Dystopian	1949	0	6	Borrowed
304	The Catcher in the Rye	J.D. Salinger	Fiction	1951	2	3	Available
305	Moby-Dick	Herman Melville	Adventure	1851	5	5	Available
306	Pride and Prejudice	Jane Austen	Romance	1813	1	4	Available
307	The Hobbit	J.R.R. Tolkien	Fantasy	1937	0	7	Borrowed
308	War and Peace	Leo Tolstoy	Historical	1869	2	5	Available
309	Hamlet	William Shakespeare	Drama	1603	4	6	Available
310	Crime and Punishment	Fyodor Dostoevsky	Crime	1866	3	4	Available

1. Create a new database called **LibraryDB.**

```
use LibraryDB
```

2. Create a collection named **Books**.

```
db.createCollection("Books")
```

3. Insert all the book records from the dataset above.

```
db.Books.insertMany([
{BookID:301, Title:"The Great Gatsby", Author:"F. Scott Fitzgerald", Genre:"Fiction",
PublishedYear:1925, CopiesAvailable:3, TotalCopies:5, Status:"Available"},
{BookID:302, Title:"To Kill a Mockingbird", Author:"Harper Lee", Genre:"Fiction",
PublishedYear:1960, CopiesAvailable:1, TotalCopies:4, Status:"Available"},
{BookID:303, Title:"1984", Author:"George Orwell", Genre:"Dystopian", PublishedYear:1949,
CopiesAvailable:0, TotalCopies:6, Status:"Borrowed"},
{BookID:304, Title:"The Catcher in the Rye", Author:"J.D. Salinger", Genre:"Fiction",
PublishedYear:1951, CopiesAvailable:2, TotalCopies:3, Status:"Available"},
{BookID:305, Title:"Moby-Dick", Author:"Herman Melville", Genre:"Adventure", PublishedYear:1851,
CopiesAvailable:5, TotalCopies:5, Status:"Available"},
{BookID:306, Title:"Pride and Prejudice", Author:"Jane Austen", Genre:"Romance",
PublishedYear:1813, CopiesAvailable:1, TotalCopies:4, Status:"Available"},
{BookID:307, Title:"The Hobbit", Author:"J.R.R. Tolkien", Genre:"Fantasy", PublishedYear:1937,
CopiesAvailable:0, TotalCopies:7, Status:"Borrowed"},
{BookID:308, Title:"War and Peace", Author:"Leo Tolstoy", Genre:"Historical", PublishedYear:1869,
CopiesAvailable:2, TotalCopies:5, Status:"Available"},
{BookID:309, Title:"Hamlet", Author:"William Shakespeare", Genre:"Drama", PublishedYear:1603,
CopiesAvailable:4, TotalCopies:6, Status:"Available"},
{BookID:310, Title:"Crime and Punishment", Author:"Fyodor Dostoevsky", Genre:"Crime",
PublishedYear:1866, CopiesAvailable:3, TotalCopies:4, Status:"Available"}
])
```

4. Increase the total copies of "**1984**" by **2** and adjust available copies accordingly.

```
db.Books.updateOne(
  { Title: "1984" },
  {
    $inc: { TotalCopies: 2, CopiesAvailable: 2 }
  }
)
```

5. Mark "**Pride and Prejudice**" as "**Reserved" if copies available are less than 2.**

```
db.Books.updateOne(
  { Title: "Pride and Prejudice", CopiesAvailable: { $lt: 2 } },
  { $set: { Status: "Reserved" } }
)
```

6. Update all books published **before 1900** to have a **new field "Classic**": true.

```
db.Books.updateMany(
  { PublishedYear: { $lt: 1900 } },
  { $set: { Classic: true } }
)
```

7. Change the **genre of "The Hobbit**" to "**Epic Fantasy**".

```
db.Books.updateOne(
  { Title: "The Hobbit" },
  { $set: { Genre: "Epic Fantasy" } }
)
```

8. Add a "**LastBorrowedDate**" field to books that are currently borrowed (set a sample date).

```
db.Books.updateMany(
  { Status: "Borrowed" },
  { $set: { LastBorrowedDate: new Date("2026-01-01") } }
)
```

9. Delete all books where **TotalCopies** is **less than 3** (outdated books removal).

```
db.Books.deleteMany(
  { TotalCopies: { $lt: 3 } }
)
```

10. Remove the "**Crime and Punishment**" book from the database.

```
db.Books.deleteOne(
  { Title: "Crime and Punishment" }
)
```

11. Remove the "**Classic"** field from books **published after 1900** (to correct the dataset).

```
db.Books.updateMany(
  { PublishedYear: { $gt: 1900 } },
  { $unset: { Classic: "" } }
)
```

12. Delete books that have been **borrowed but have 0 total copies left**.

```
db.Books.deleteMany(
  {
    Status: "Borrowed",
    TotalCopies: 0
  }
)
```